Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Preclinical Process Sensitivity and Reproducibility Assessment for CB-to-Interim ECFC Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Cell and Tissue Therapeutics	R&D/Engineering Services	Production Management/ Quantitative Analysis Technique
Concept of Proposed Technology	The proposed technology is a **preclinical process sensitivity and reproducibility assessment framework** designed to experimentally identify and characterize **sources of batch-to-batch variability** occurring during the **CB-to-Interim stage** of ECFC manufacturing. Rather than developing new cell therapies, optimizing protocols, or generating clinical or regulatory submission data, this project focuses on **non-GLP preclinical execution** to evaluate **repeatability and sensitivity** of early-stage manufacturing processes. Specifically, the framework integrates: • **Repeatability assessment** through 3–5 identical CB-to-Interim process runs to quantify baseline variability • **Process sensitivity evaluation** by controlled perturbation of a single process factor within predefined windows to identify variability amplification points • **Process domain structuring and risk prioritization**, translating experimental observations into a **Process Sensitivity & Variability Map**			

	• **Executable preclinical data structuring**, enabling traceable, non-regulatory datasets suitable for expert review and downstream planning This approach establishes a **defensible preclinical manufacturing knowledge base**, reducing uncertainty prior to transition into MCB establishment, downstream manufacturing, or future clinical development activities.
Rationale for Joint R&D Project	Regenerative medicine and cell-based therapies frequently encounter significant challenges **before** entering downstream development stages, not due to lack of scientific innovation, but because of **unresolved variability in early-stage manufacturing processes**. In particular, the **CB-to-Interim stage** represents a critical transition point where biological heterogeneity, handling conditions, and early process decisions can lead to substantial **batch-to-batch variability**. Once variability is introduced at this stage, it propagates downstream, undermining reproducibility and increasing risk in subsequent manufacturing or development steps. Many SMEs attempt to advance toward MCB establishment or clinical manufacturing without first establishing a **data-supported understanding of early-stage process repeatability and sensitivity**. This often results in inefficiencies, unanticipated failures, and difficulty aligning with global research partners who require clear evidence of process robustness prior to further collaboration. This joint R&D project addresses this gap by collaborating with a global research institute possessing expertise in **translational manufacturing and preclinical process analysis**. Through limited, non-GLP preclinical execution, the project experimentally identifies and prioritizes **sources of variability** within the CB-to-Interim process, translating empirical observations into a structured **Process Sensitivity & Variability Map**. By establishing a shared, experimentally grounded understanding of early-stage process behavior, the project enables more informed decision-making before committing to resource-intensive downstream development, thereby reducing technical uncertainty and strengthening the foundation for future international joint R&D.
Key Words	1. **Regenerative Medicine** 2. **Preclinical Process Sensitivity** 3. **Manufacturing Reproducibility** 4. **Batch-to-Batch Variability** 5. **International Joint R&D (Pre-Planning)**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (4)
R&D Objectives	The objectives of this project are as follows: 1. To **define and structure key CB-to-Interim process steps** in ECFC manufacturing, focusing on early-stage transitions that contribute to outcome variability. 2. To **experimentally assess process repeatability** through multiple identical CB-to-Interim runs and quantify baseline batch-to-batch variability. 3. To **identify and characterize process sensitivity** by evaluating the impact of controlled perturbation of a single early-stage process factor on outcome distribution. 4. To **prioritize manufacturing-related technical risks** at the CB-to-Interim stage and define data-supported gate criteria for downstream transition.			
R&D Target Specifications	The project aims to deliver the following tangible outputs: • A **CB-to-Interim Repeatability Assessment Report**, summarizing variability metrics derived from 3–5 identical process executions • A **Process Sensitivity & Variability Map**, identifying early-stage process factors that			

	amplify batch-to-batch variability • A **Manufacturing Risk Prioritization Table**, ranking variability drivers by impact and controllability • A **Preclinical Transition Gate Definition Document**, outlining minimum consistency requirements prior to MCB establishment or downstream development • A **non-GLP executable data package**, ensuring traceability and internal reproducibility without disclosure of proprietary implementation details
R&D Plan	**Phase 1: CB-to-Interim Process Structuring and Repeatability Assessment** This phase defines and structures the CB-to-Interim process at a conceptual level and evaluates baseline repeatability. The identical CB-to-Interim workflow is executed 3–5 times to quantify batch-to-batch variability in early outcomes such as Interim transition success and colony emergence consistency. No optimization, clinical testing, or regulatory submission activities are performed. **Phase 2: Process Sensitivity Evaluation (Single-Factor Perturbation)** This phase assesses process sensitivity by intentionally perturbing a single Tier 1 early-stage process factor within predefined windows across 2–3 levels. The objective is to determine whether controlled variation increases outcome distribution spread rather than to identify optimal conditions. All perturbations are limited to CB-to-Interim stages and conducted at a non-GLP preclinical level. **Phase 3: Variability Interpretation and Risk Prioritization** Experimental observations from repeatability and sensitivity assessments are integrated to interpret variability patterns. Process factors are classified based on their contribution to batch-to-batch variability and controllability, resulting in a prioritized manufacturing risk profile specific to the CB-to-Interim stage. **Phase 4: Data Structuring and Transition Gate Definition** This phase structures experimental data into a traceable, non-regulatory electronic format and defines objective transition gate criteria required before proceeding to MCB establishment or downstream manufacturing. The framework supports internal decision-making and future international joint R&D planning without exposing proprietary process details.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The global research institute will: • **Co-design and execute non-GLP preclinical studies** focused on CB-to-Interim process repeatability and sensitivity assessment • Support **structuring of early-stage process domains** and experimental frameworks for identifying sources of batch-to-batch variability • Participate in **interpretation of experimental data**, with emphasis on translational manufacturing perspectives rather than clinical or regulatory endpoints • Contribute to the development of a **Process Sensitivity & Variability Map** and a prioritized manufacturing risk profile specific to the CB-to-Interim stage • Assist in defining **data-supported transition gate criteria** required prior to MCB establishment or downstream manufacturing activities The institute is expected to have demonstrated expertise in **preclinical process execution, tissue engineering, and translational manufacturing research**, enabling meaningful participation in experimental execution and analysis without engaging in clinical testing or regulatory submission activities.

Expected R&R of Korean SME Partner	The Korean SME partner will: • Provide **CB-derived biological materials and internal process workflows** required for CB-to-Interim execution, while retaining ownership of proprietary formulations and SOPs • Lead **hands-on execution of CB-to-Interim processes**, including repeatability and sensitivity experiments, under non-GLP conditions • Generate and manage **primary experimental data**, ensuring traceability, consistency, and internal reproducibility • Collaborate with the global research institute in **data analysis and risk interpretation**, incorporating external perspectives into internal decision-making • Integrate project outcomes into **subsequent internal R&D planning**, including MCB establishment, manufacturing strategy refinement, or follow-on international joint R&D initiatives

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** This project will deliver experimentally grounded, non-GLP preclinical outcomes that clarify how early-stage process behavior influences reproducibility in ECFC manufacturing. Specifically, the project will generate: • A CB-to-Interim Repeatability Assessment, quantifying baseline batch-to-batch variability through 3–5 identical process executions • A Process Sensitivity & Variability Map, supported by experimental data, identifying early-stage process factors that disproportionately amplify outcome variability • A Manufacturing Risk Prioritization Profile for the CB-to-Interim stage, enabling data-driven identification of high-impact variability drivers • Objective transition gate criteria defining minimum consistency requirements prior to progression into MCB establishment or downstream manufacturing steps • A traceable, non-regulatory preclinical data package suitable for internal decision-making and expert review without disclosure of proprietary implementation details Together, these outcomes establish a defensible preclinical manufacturing knowledge base that reduces uncertainty at the most failure-prone early stage of development. **Expected Impact and Utilization** The impact of this project extends beyond immediate experimental results by enabling more reliable progression toward downstream development activities. Project outcomes will be utilized to: • De-risk subsequent internal R&D activities, including MCB establishment and downstream manufacturing planning, by addressing early-stage variability before scale or complexity increases • Support follow-on international joint R&D programs, providing empirically justified process understanding rather than assumption-based planning • Facilitate technical alignment discussions with global research partners, grounded in reproducibility and process behavior rather than isolated performance claims • Strengthen manufacturing readiness narratives for future investment, partnership, or technology transfer discussions by demonstrating disciplined preclinical process control • Rather than producing short-term performance claims, this project creates a foundational, execution-based understanding of early manufacturing variability, enabling sustainable and credible advancement of regenerative medicine technologies toward long-term global development pathways.

6. References

(1) International Council for Harmonisation (ICH) Guidelines

(2) Global Regulatory Frameworks for Regenerative Medicine